

L^AT_EX export hooks explained

SCS

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Contents

Overview

This document explains how Org → L^AT_EX/PDF export is extended for SCS checklists and similar documents. The pieces work together:

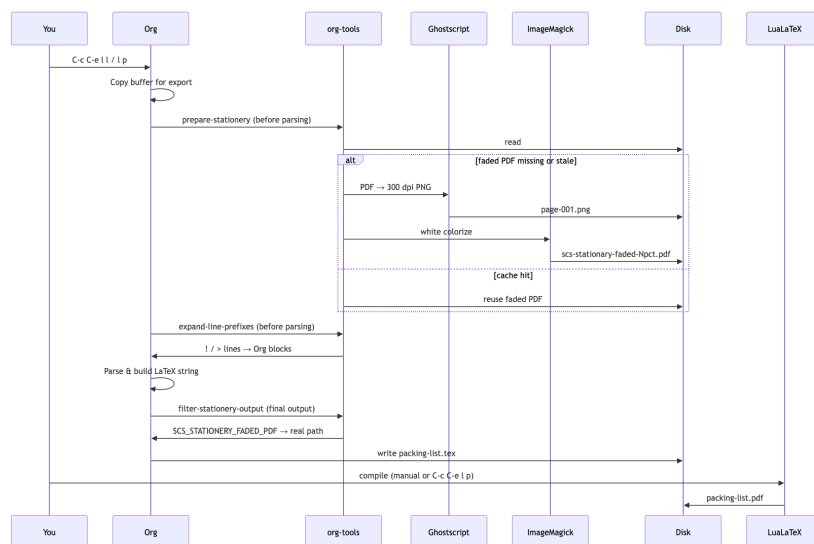
Component	File	Role
Line-prefix syntax	<code>org-tools.el</code>	! / => shorthands → center / quote blocks
Faded letterhead	<code>org-tools.el</code>	Reads Org keywords, builds faded PDF, patches <code>.tex</code>
Canon fonts	<code>scs-fontspec.tex</code>	Simoncini body, Adobe Garamond small caps & headings
Org document	e.g. <code>packing-list.org</code>	Keywords, <code>#+LATEX_HEADER</code> , body text

Enable once in your init file:

```
(with-eval-after-load 'org
  (require 'org-tools))
```

The export pipeline (big picture)

When you run `C-c C-e l l` (L^AT_EX export) or `C-c C-e l p` (PDF), Org does not simply “save the buffer as `=.tex=`”. Several hooks run in a fixed order.



Hook registration (in org-tools.el):

Hook	Function	When
org-export-before-parsing-functions	org-tools-prepare-stationery	Before C
org-export-before-parsing-functions	org-tools-expand-line-prefixes	Same ph
org-export-filter-final-output-functions	org-tools-filter-stationery-output	After ful

Stationery runs first (prepends to the hook list after line-prefixes is registered, so it executes before prefix expansion).

Line-prefix syntax (! and >)

Implemented in /Users/scs/.config/emacs/lisp/org-tools.el.

When it runs: org-export-before-parsing-functions → org-tools-expand-line-prefixes, before Org parses the export buffer.

Toggle: M-x org-tools-line-prefixes-mode / M-x org-tools-line-prefixes-status

Centered lines (!)

Consecutive lines starting with ! at column 0 become a center block. Optional style tokens after ! apply to that line only.

```
!sc large Packing checklist
! (summer - warm season)
```

L^AT_EX export (conceptually):

```
#+begin_center
@@latex:{\large\textsc{Packing checklist}}@@
```

```
(summer - warm season)
#+end_center
```

Quote lines (>)

```
> Nothing is compulsory.
> Some items are optional.
```

becomes:

```
#+begin_quote
Nothing is compulsory.

Some items are optional.
#+end_quote
```

Style modifiers (center lines only, space-separated, case-sensitive):

Token	L ^A T _E X effect
sc, smallcaps	#ERROR
tiny ... huge, HUGE	\tiny ... \huge
10pt, 12.5pt, etc.	\fontsize{...}{...}\selectfont

Examples:

```
!sc large Main title
!sc footnotesize typeset at tapoban -- 4 july 63 sc
!14pt Subtitle
! Plain centred line
```

Rules

- Prefixes start at column 0.
- A blank line ends the current run.
- ! and => runs do not mix.

- Lines inside `#+begin_src`, `#+begin_export`, `#+begin_example`, `#+begin_verse` are left untouched.

Footer layout helpers

Push a credit line to the bottom of the page:

```
#+begin_export latex
\vfill
#+end_export
```

```
!sc footnotesize typeset at tapoban on 4 july 63 sc -- sri chinmoy society press
!sc footnotesize [letterhead colours dimmed on purpose]
```

Use `\vspace{-0.5em}` after `\vfill` if you want the line closer to the bottom margin.

Canon fonts (scs-canon)

File: `/Users/scs/fonts/scs-canon/scs-fontspec.tex`

All faces live in one directory so fontspec auxiliary fonts (`SmallCapsFont`, `BoldFont`, etc.) can share a single Path. Org files pull it in with:

```
#+LATEX_HEADER: \usepackage{microtype}
#+LATEX_HEADER: \input{/Users/scs/fonts/scs-canon/scs-fontspec.tex}
```

Load `microtype` before `scs-fontspec.tex` so `\textsc` is reinstalled correctly after `microtype` loads.

Faces configured

Macro / role	Font file	Notes
Body (<code>\setmainfont</code>)	<code>SimonciniGaramondStd</code>	Regular, bold, italic
<code>\textsc</code>	<code>AGaramondPro-Regular + Letters=SmallCaps</code>	Simoncini has no smcp gly
<code>\runheaderfont</code>	<code>AGaramondPro</code> family	Running headers
<code>\runheadertext{...}</code>	wraps <code>\scs@smallcaps</code>	Safe inside <code>\fancyhead</code>

Why not `SmallCapsFont={AGaramond...}` alone?

`SmallCapsFont` selects the **file** for the NFSS `sc` shape but does not enable OpenType `smcp`. The working approach uses an explicit `\fontspec` call with `Letters=SmallCaps`:

`\fontspec[Path=..., Letters=SmallCaps, Ligatures=TeX]{AGaramondPro-Regular}{#1}`

Important details (learned the hard way):

Issue	Cause	Fix
No visible small caps	<code>smcp</code> not enabled	<code>Letters=SmallCaps</code> on
Text starting with <code>[</code> vanishes	<code>[...]</code> parsed as fontspec options	Wrap text in <code>{#1}</code> after
<code>--</code> stays as two hyphens	HarfBuzz path skips $\TeX$ ligatures	<code>Ligatures=TeX</code> on smal
Header <code>[packing checklist]</code> broken	<code>#1</code> in hook macro body	Use <code>##1</code> in <code>\scs@insta</code>
<code>\textsc</code> broken after microtype	Hook passes args into <code>#1</code>	<code>\AddToHook{package/m</code>

Swap faces by editing the `\scs@...` variables at the top of `scs-fontspec.tex`.

Faded letterhead (stationery)

Vector letterhead (`scs-stationary.pdf`) is PDF/X CMYK. $\LaTeX$ `\includegraphics[opacity=0.5]` had no visible effect. The working approach pre-fades via Ghostscript + ImageMagick and includes the rasterised PDF as background (`eso-pic`).

Org file keywords

`#+SCS_STATIONERY: /Users/scs/Documents/scs-stationary.pdf`

`#+SCS_STATIONERY_FADE: 27`

`#+LATEX_HEADER: \AddToShipoutPictureBG{\AtPageLowerLeft{\includegraphics[width=\paperw`

Keyword	Meaning
<code>#+SCS_STATIONERY:</code>	Source letterhead PDF (default: <code>~/Documents/scs-stationary.pdf</code>)
<code>#+SCS_STATIONERY_FADE:</code>	Percentage of original colour kept (27 = quite faint)
<code>SCS_STATIONERY_FADED_PDF</code>	Placeholder in <code>#+LATEX_HEADER</code> ; replaced in final <code>.tex</code>

Do **not** use `@@SCS_STATIONERY@@` — Org treats `@@...@@` as export snippets.

Cached output path

`{source-dir}/{basename}-faded-{N}.pct.pdf`

Example: `/Users/scs/Documents/scs-stationary-faded-27pct.pdf`

Changing FADE from 27 to 20 creates a separate cached file; old percentages remain on disk.

Fade math

ImageMagick colorize% = 100 FADE%

FADE 27 → colorize 73% white (27% colour remains)

FADE 50 → colorize 50% white

Step 2 behind the scenes (raster fade)

When `org-tools-prepare-stationery` runs:

1. **Read keywords** from the export buffer (first 32 KiB), via regex on `#+SCS_STATIONERY:` and `#+SCS_STATIONERY_FADE:`.
2. **Compute paths** — source PDF and faded output `...-faded-Npct.pdf`.
3. **Skip or rebuild** — rebuild only if faded file missing or older than source.
4. **Run three external commands** in a temp directory (`call-process`, not a shell script on disk):

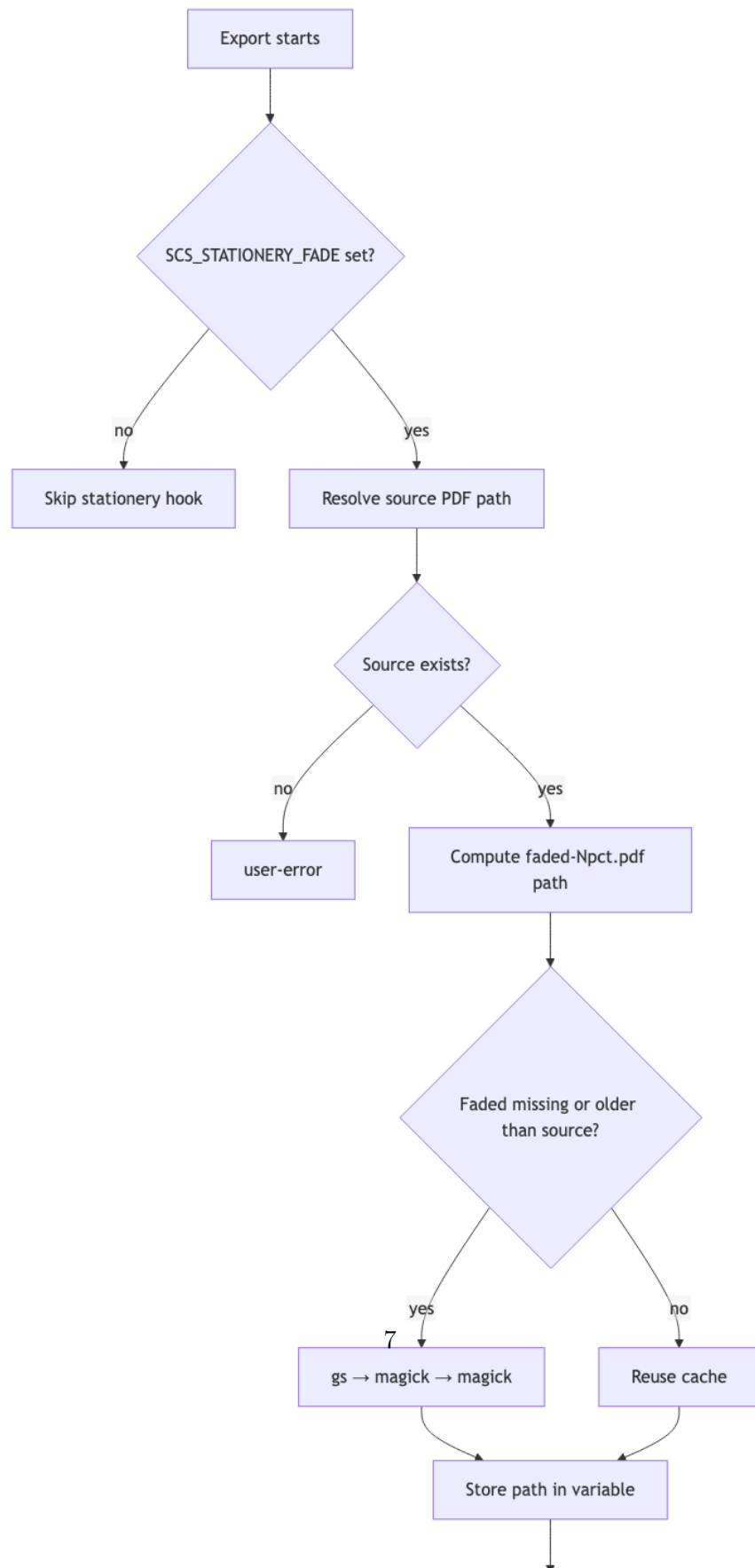
Step	Program	Arguments (conceptually)	Result
A	gs	png16m, -r300, source PDF	page-001.png
B	magick	white background, -colorize N%	faded.png
C	magick	300 dpi, zip compress	scs-stationary-faded-27pct.pdf

1. **Delete temp directory** (`unwind-protect`).
2. **Store path** in `org-tools--stationery-faded-path` for this export.

Elisp entry points:

Function	Purpose
<code>org-tools-prepare-stationery</code>	Hook: ensure faded PDF exists
<code>org-tools-ensure-stationery-faded</code>	Read keywords, cache logic, maybe rebuild
<code>org-tools--regenerate-stationery-faded</code>	Run gs + magick pipeline
<code>org-tools-filter-stationery-output</code>	Replace placeholder in final <code>.tex</code> string
<code>org-tools-regenerate-stationery</code>	M-x — rebuild without exporting

Regeneration decision



Manual equivalent (what the `Elisp` shim runs):

```
SRC=/Users/scs/Documents/scs-stationary.pdf
OUT=/Users/scs/Documents/scs-stationary-faded-27pct.pdf
FADE=27
COLORIZE=$((100 - FADE))    # 73

gs -q -dNOPAUSE -dBATCH -dNOSAFAER \
  -sDEVICE=png16m -r300 \
  -sOutputFile=/tmp/page-%03d.png "$SRC"

magick /tmp/page-001.png -background white -alpha remove \
  -fill white -colorize ${COLORIZE}% /tmp/faded.png

magick /tmp/faded.png -units PixelsPerInch -density 300 \
  -compress zip "$OUT"
```

Placeholder substitution happens on the **final `LATEX` string**, not in the export buffer (Org assembles `#+LATEX_HEADER` separately). Substitution uses `replace-match` with `fixedcase` so paths like `/Users/scs/...` are not corrupted (plain `replace-regexp-in-string` uppercased paths because of `\U` escape rules in replacement templates).

Example `packing-list.org` skeleton

```
#+TITLE: Packing checklist
#+SCS_STATIONERY: /Users/scs/Documents/scs-stationary.pdf
#+SCS_STATIONERY_FADE: 27
#+LATEX_COMPILER: lualatex

#+LATEX_HEADER: \usepackage{eso-pic}
#+LATEX_HEADER: \usepackage{geometry}
#+LATEX_HEADER: \geometry{...}
#+LATEX_HEADER: \AddToShipoutPictureBG{...{SCS_STATIONERY_FADED_PDF}}
#+LATEX_HEADER: \usepackage{microtype}
#+LATEX_HEADER: \input{/Users/scs/fonts/scs-canon/scs-fontspec.tex}
#+LATEX_HEADER: \fancyhead[C]{\runheadertext{[packing checklist]}}

... body ...
```



```
#+begin_export latex
\vfill \vspace{-0.5em}
#+end_export
```

```
!sc footnotesize typeset at tapoban on 4 july 63 sc -- sri chinmoy society press
!sc footnotesize [letterhead colours dimmed on purpose]
```

Requirements and troubleshooting

Requirements

- Emacs 29.1, Org 9.0
- LuaLaTeX (`#+LATEX_COMPILER: lualatex`)
- `fontspec`, `eso-pic`, `fancyhdr`, `microtype` (via T_EX Live)
- `gs` (Ghostscript) and `magick` (ImageMagick) on `PATH` for stationery fade
- Fonts in `/Users/scs/fonts/scs-canon/`

Common issues

Symptom	Likely cause	What to do
! lines not expanded	<code>org-tools</code> not loaded	(<code>require 'org-tools</code>) in <code>init</code>
No small caps	Old <code>scs-fontspec.tex</code>	Reload; check <code>Letters=SmallCaps</code>
Line with [missing	<code>fontspec</code> ate [as options	Ensure <code>{#1}</code> wrapper in <code>scs-font</code>
-- not an en-dash in !sc	Missing <code>Ligatures=TeX</code>	Update <code>scs-fontspec.tex</code>
Header brackets missing	Fragile <code>\textsc</code> in <code>\fancyhead</code>	Use <code>\runheadertext{...}</code>
Wrong stationery in PDF	Stale cache	Touch source PDF or M-x <code>org-to</code>
Export error “gs not found”	Ghostscript missing	Install Ghostscript
Path all UPPERCASE in .tex	Old substitution bug	Update <code>org-tools.el</code> (use <code>repla</code>

Useful commands

Command	Action
C-c C-e l l	Export to L ^A T _E X file
C-c C-e l p	Export to PDF (export + compile)
M-x <code>org-tools-line-prefixes-status</code>	Check ! / => hook
M-x <code>org-tools-regenerate-stationery</code>	Rebuild faded letterhead from buffer keywords

File index

Path	Description
<code>~/.config/emacs/lisp/org-tools.el</code>	Line prefixes + stationery hooks
<code>~/.config/emacs/lisp/doc/latex-export-hook-explained.org</code>	This document
<code>/Users/scs/fonts/scs-canon/scs-fontspec.tex</code>	Canon font setup
<code>/Users/scs/fonts/scs-canon/*.otf</code>	Simoncini + Adobe Garamond f
<code>~/Documents/scs-stationary.pdf</code>	Full-strength letterhead source
<code>~/Documents/scs-stationary-faded-Npct.pdf</code>	Cached faded letterhead

Viewing Mermaid diagrams in Org

This file uses `#+begin_src mermaid` blocks. They render when you evaluate them (`C-c C-c` on the block) or export the document.

Setup (already in `init.el`):

- `ob-mermaid` (`el-get`) + `mermaid` in `org-babel-load-languages`
- `mmdc` from `npm install -g @mermaid-js/mermaid-cli`
- Config: `~/.config/mermaid/config.json` (`htmlLabels: false` for Emacs SVG)

Commands

Action	Keys
Render one block	<code>C-c C-c</code> on the block
Export whole doc	<code>C-c C-e h o</code> (HTML) or <code>C-c C-e l p</code> (PDF via $\text{\LaTeX}$)

Diagram PNGs are written beside this file (`export-pipeline.png`, `stationery-regenerate.png`).

History / design notes

- `scrartcl` was avoided for title placement — Org batch export did not register the class without extra Elisp; `titling` + custom `\maketitle` on `article` was used instead.
- **TikZ opacity** on vector stationery worked but was dropped in favour of pre-faded assets for predictable print output.

- **SmallCapsFont=** in a single `\setmainfont` block works when all faces share one directory (**scs-canon**); cross-directory nested **SmallCapsFont**[Path=...]= failed on T_EX Live 2026 fontspec.
- Stationery placeholder moved from `@@. . .@@` to `SCS_STATIONERY_FADED_PDF` because Org interprets double-@ as export snippets.